TruMarkZ

VERIFY • TRUST • TRANSFORM

Blockchain-Backed Verification & Credentialing Platform

Product Flow & Build Document — Prepared by Development Team — April 2026

TruMarkZ is a single platform for verifying **anything and anyone** — workers, products, services, and organisations — issuing **tamper-proof digital credentials** backed by the Dhiway blockchain, and making those credentials searchable, shareable, and verifiable by anyone.

For Organisations	For Individuals	Blockchain	Skill Trees	Registry
Verify workers & products in bulk	Build a skill tree credential	Dhiway CORD — tamper-proof	Visual digital resume	Search & download reports

Platform	Android App (Flutter) + Web Portals
Blockchain	Dhiway CORD Network (dhiway.com)
Verification	API-based (auto) + Human verifiers (manual)
Who Uses It	Organisations, Individuals, Super Admin, Public
Cloud	Amazon Web Services — India Region
Duration	10 Weeks — Build + Test + Launch
Team	4 people (PM + Flutter + Web + Backend)

1. What is TruMarkZ?

TruMarkZ is not just a background check tool. It is a complete Trust Infrastructure Platform — one system that verifies anything (workers, products, services) issues tamper-proof digital credentials, and makes them searchable and shareable.

For Organisations & Businesses

→ Verify 200 drivers? Upload an Excel and TruMarkZ verifies each one.

→ Organic fertiliser company? Get a digital certificate proving no harmful chemicals used.

→ Silk manufacturer? Credential proves no silkworms were killed — buyers can scan before purchasing.

→ Any industry, any product, any service can have a verified credential.

For Individuals

→ An engineer with 2 years of experience can create their own verified credential.

→ A nurse can build a skill tree credential to send to hospitals and recruiters.

→ A fresher can show their degree, courses, and projects — all verified on blockchain.

→ Anyone can be both an 'organisation' and an individual on TruMarkZ.

For Employers & Public

→ A traveller can scan a driver's QR code to check their verified record before getting in.

→ A hospital can search for a nurse and view her credentials before hiring.

→ A logistics company can verify a driver's background by paying ₹5–₹10.

→ Anyone with a QR code or link can verify — no app download needed.

2. Organisation Flow — Step by Step

An 'Organisation' can be anything: a transport company, a nursing college, a manufacturer, a hospital, an association with thousands of drivers, or even an individual who wants a verified credential for themselves.

STAGE 1 — ORGANISATION REGISTRATION & VERIFICATION

1	**Register Organisation** Enter: Organisation name, Industry (dropdown), Address, GST number, Business Registration number, Official email (OTP verified). Can also register as an Individual.
2	**Organisation Verification by Super Admin** Once registered, the super admin verifies the organisation before it can use the platform. Verification uses GST API check + official email OTP. The organisation cannot upload data until approved. Approval should be fast — target under 24 hours.
3	**Onboarding Confirmation** Organisation receives email confirmation that their account is live. They can now log in and configure their verification plan.

STAGE 2 — VERIFICATION PLAN CONFIGURATION

4	**Select Industry** Choose from: Transport, Healthcare, Education, Manufacturing, Security, Agriculture, Products/Services, Others.
5	**Select Verification Checks** Tick which checks are needed. Each check is either API-based (automatic) or Human-based (manual). System clearly shows which is which. Examples: Identity (API) | Address (Manual) | Police Clearance (Manual) | Education (API) | Employment (API) | Criminal Record (Manual) | Product/Compliance (Manual).
6	**See Per-Unit Cost** No budget slider — instead, the system shows a clean per-unit cost breakdown: Identity Check — ₹X | Education Check — ₹Y | Total per person/product — ₹X+Y. Organisation agrees to the cost before proceeding.
7	**Permission-Based Access Setting** Organisation chooses: should this credential be publicly searchable, or permission-based? For sensitive profiles (e.g. nursing graduates, engineers) — permission required before anyone can search. For general profiles (e.g. drivers) — open search. Permission triggers a WhatsApp/email message to the person being verified for consent.

Our Suggestion — Permission & Consent Flow

→ When permission is required: system sends a WhatsApp message or email to the worker/individual with an 'I Agree' button. Cost of this message (~₹1) is included in the per-unit fee charged to the organisation.

→ When permission is NOT required (e.g. bulk driver verification): no message sent, verification proceeds immediately.

→ For bulk batches: we recommend a pre-consent clause in the organisation's agreement with their workers — so no individual messages are needed for every record.

→ OTP-based consent works for individual uploads; for bulk (1,000+ records), WhatsApp broadcast with 'I Agree' link is our recommended approach.

STAGE 3 — DATA UPLOAD (BULK & SINGLE)

Bulk Upload Flow

8	**Download Excel Template** System generates a customised Excel template based on the checks selected. The template is a superset of all fields needed across all selected checks. E.g. if Education + Criminal + Identity selected: template includes Name, DOB, Address, ID Number, College Name, Graduation Year, Photo filename. Image column contains the filename — a separate ZIP of photos is uploaded alongside.
9	**Fill & Upload Excel + Photo ZIP**

	Organisation fills the Excel (all worker/product details). Photos are placed in a ZIP file with filenames matching the Excel column. Both files uploaded together. System previews all entries — organisation can review before confirming.
10	**Batch Created** Organisation names the batch (e.g. 'Driver Verification Q1 — 200 records'). System creates one verification task per check per record. Processing begins immediately.

Single / Individual Upload Flow

11	**Fill Individual Form** For single records: organisation fills an online form directly in the app (same fields as the Excel template). Photo uploaded from device or taken via camera.
12	**Submit & Track** Single record submitted. Tracking works the same way as bulk — shows status per check in real time.

Our Suggestion — Image Handling (raised by client)

→ Images are collected in the Excel as a filename reference (e.g. ravi_kumar.jpg). A ZIP of all photos is uploaded alongside the Excel — system matches by filename.

→ For individual upload: photo is uploaded directly via the app (camera or gallery picker).

→ All photos are stored in AWS S3 (encrypted, private). A secure link is generated when the credential is created.

→ For bulk generation (10,000 credentials): each credential auto-pulls the matching photo from S3 using the filename from the Excel row.

STAGE 4 — VERIFICATION PROCESSING	
13	**API Checks Run Automatically** For API-based checks (Aadhaar, PAN, Driving Licence, Education via NAD, Employment via EPFO): system calls government APIs instantly. Results stored in the unified verification record. No human involvement needed.
14	**Human Verification Tasks Assigned** For manual checks (Address visit, Police clearance, Product compliance): Super Admin defines the verifier (third-party agency) for this check type. System automatically emails the verifier: scope of work, list of records (as Excel attachment), SLA timeline, and pricing (70% of the per-check cost charged to the org).
15	**SLA Monitoring** Super Admin dashboard shows live status per batch: which verifier, how many records, SLA deadline, % completed. If only 30% of time remains and batch is incomplete — automatic alert email sent to verifier. Super Admin sees a red alert on the dashboard.
16	**Results Collected** API results come in automatically. Human verifier uploads completed Excel + evidence photos back to the platform. All results stored in the unified record — same format regardless of check type.

STAGE 5 — FINAL REPORT GENERATION	
17	**Unified Verification Report Created** Once ALL checks are complete for a record, the system auto-generates one unified report combining API results + human verifier results. Available as PDF download and live digital view.
18	**Organisation Reviews** Organisation can view the full report per record. If anything looks wrong, they can raise a dispute. Super Admin resolves disputes.

3. Credential Creation — Templates & QR

Credentials are NOT just for human workers. They work for products (organic fertiliser, silk, electric vehicles), services (driver safety), and individuals (skill tree resume). Different templates exist for each use case.

Certificate Templates

The client confirmed 6–7 template types. Each template defines which fields appear on the front of the digital ID card. The organisation picks a template when generating credentials.

#	Template	Designed For	Key Fields Shown on Card
T1	**Workforce / Driver ID**	Transport companies, security firms, logistics	Photo, Name, Designation, Service No., Licence No., Verified Checks, QR
T2	**Healthcare / Nurse**	Hospitals, nursing colleges, healthcare orgs	Photo, Name, Qualification, Registration No., Issuing Body, QR
T3	**Education / Student**	Colleges, universities, training institutes	Photo, Name, Course, Institution, Year, Credential ID, QR
T4	**Product / Compliance**	Manufacturers, agriculture, organic goods, e-commerce	Product Name, Batch No., Compliance Type, Issuing Org, Verified By, QR
T5	**Service / Professional**	Individual professionals, freelancers, engineers	Photo, Name, Skill/Service, Years of Experience, Verified By, QR
T6	**Skill Tree Credential**	Individuals building a verified digital resume	Photo, Name, Skills with verification nodes, Credential IDs, QR

Field Selection — Choosing What Goes on the Card

The organisation's Excel may have 10+ fields, but only 5–6 should appear on the certificate face. Here is how it works:

1	**Pick a Template** Organisation selects the template that matches their use case (e.g. Workforce/Driver, Healthcare, Product).
2	**Map Fields** A simple drag-and-drop or checkbox screen shows all available fields from their Excel. Organisation ticks which 5–6 fields should appear on the credential face. Remaining fields stay in the background report only.
3	**Preview** See exactly what the credential will look like — with a real worker's data — before generating for all records.
4	**Approve & Generate** Click Approve. System generates credentials for ALL records in the batch at once — each with the correct photo, name, and details pulled automatically from the matching Excel row.

Our Suggestion — Bulk Credential Generation for 10,000 Records

→ Each credential is generated individually but the process runs as a background batch job — organisation does not wait.

→ Photo is pulled from S3 using the filename from the Excel row. If a photo is missing for a record, that record is flagged for manual attention.

→ Credentials are generated one by one in the background. Organisation gets a notification when all are ready.

→ The Dhiway blockchain write happens once per credential — hashing the credential data. This is also batched to reduce costs.

The Digital Credential — What It Looks Like

On the Front of the Card

- Organisation logo + name
- Photo (if applicable)

Behind the QR (Full Report)

→ Full unified verification report

- Verified name
- Unique Credential ID — e.g. TM-TRN-2026-00394
- ✓ Selected fields (5–6 chosen by organisation)
- Issue date + Valid until date
- QR Code — scan to verify instantly
- Dhiway blockchain hash (first 8 characters shown)

→ API check results (Aadhaar, PAN, DL, Education, Employment)
→ Human verifier results (Address, Police, Compliance)
→ Evidence photos from field agents
→ Dhiway CORD blockchain proof: hash, timestamp, network, integrity status
→ Credential status: VALID / EXPIRED / REVOKED
→ Permission-based: some sections gated behind ₹5–₹10 payment

4. QR Scanning & Public Verification Flow

Anyone can scan the QR on a TruMarkZ credential — from any phone camera. No app download. No login. Results appear in under 2 seconds.

1	**Scan the QR** Open phone camera (or TruMarkZ app). Point at QR on the credential. Works from a printed card, phone screen, WhatsApp, email attachment, or PDF.
2	**See the Credential** Instantly shows: photo, name, credential ID, issuing organisation, and a clear VERIFIED (green) / EXPIRED (amber) / REVOKED (red) badge.
3	**See Verification Summary** Breakdown of each check: Identity ✓ Address ✓ Police ✓ Education ✓. Shows when each check was done and by whom.
4	**Public vs Paid Access** Basic credential info is free. Full detailed reports (e.g. police clearance document, address evidence photos) require a ₹5–₹10 payment via Razorpay. After payment, report downloads instantly.
5	**Dhiway Blockchain Proof** At the bottom: transaction hash, Dhiway network name, timestamp, and 'Report Integrity: ✓Verified ✓Not Modified'. Tamper-proof — mathematically guaranteed.

Real-World Example from the Transcript:

A woman traveller gets into a cab. The driver shows his TruMarkZ credential on his phone. She scans the QR in 1 second. She sees his photo, name, verified driving licence, clean police record, and identity confirmation — all on her phone. She knows she is safe.

An e-commerce buyer wants to purchase organic silk online. They tap the QR on the product listing. They see a TruMarkZ credential confirming no silkworms were harmed, organic sourcing was verified by a third-party agent, and the credential is blockchain-recorded.

5. Super Admin — Control Layer

The Super Admin is a person from the TruMarkZ company team. They are the backbone of the platform — controlling verifiers, monitoring batches, resolving disputes, and ensuring everything runs on time.

Responsibility	What the Super Admin Does
Verify New Orgs	Review and approve newly registered organisations (using GST verification). Block suspicious registrations. Target: approve within 24 hours.
Define Verifier Pricing	Set the cost per check type (e.g. Education = ₹100/check). Verifiers receive 70% of this. Platform keeps 30%.
Assign Verifiers	For each manual check type + location, define which third-party agency (verifier) handles it. Their email ID is pre-set in the system.
Monitor Batches	Dashboard shows all active batches: organisation name, verifier assigned, total records, SLA deadline, % complete. Red alert when 30% of time remains.
SLA Alerts	System auto-sends reminder email to verifier when 30% of SLA time remains. Super Admin also gets the alert and can intervene.
Dispute Resolution	If an organisation or verifier raises a dispute about a result, Super Admin reviews evidence and makes a final ruling.
Activate Projects	After verifiers confirm a job, Super Admin clicks 'Activate Project' — this triggers the organisation's notification that verification has begun.
Platform Health	Monitor API success rates, failed verifications, payment status, and overall system health from a single dashboard.

6. Blockchain Credential Registry — Dhiway

> *All credentials are stored on the Dhiway CORD Network — a blockchain built specifically for verifiable credentials and digital identity in India. It follows global W3C standards. The CTO of Dhiway (Mark Studio) will be introduced to the development team to set up the integration.*

Component	How It Works
What Gets Stored	A SHA-256 hash of the credential data (name + checks + org + date). No personal data is stored on the blockchain — only the fingerprint. This protects privacy while proving authenticity.
Credential Issuance	Each approved credential triggers one blockchain write to Dhiway. The transaction hash is embedded in the QR code and displayed on the credential.
Mark Studio Templates	Dhiway's Mark Studio platform allows custom credential templates. The development team will work with Dhiway CTO to configure templates for each of our 6 credential types.
Immutability	Once written to Dhiway, a credential hash cannot be changed. If a credential is revoked, a revocation entry is added — the original hash remains, proving the full history.
Revocation	Organisation admin can revoke a credential at any time. Revocation is written to Dhiway's revocation registry. QR scan immediately shows REVOKED.
Bulk Writing	For bulk credential generation (e.g. 10,000 drivers), blockchain writes are batched to reduce transaction costs. Each write is queued and processed in the background.
Public Registry Search	Anyone can search by: Name, Date of Birth, Unique Credential ID, Licence Number. Basic result (name + status) is free. Full report download costs ₹5–₹10 (paid via Razorpay).
Permission Control	For sensitive profile types (nurses, engineers, semiconductor workers): a consent message is sent before their record is searchable. They can enable or disable their public visibility at any time.

7. Skill Tree — Verified Digital Resume

This is the visual intelligence layer of TruMarkZ — but simplified to just the Skill Tree as confirmed by the client. No learning modules or upskilling marketplace in Phase 1. The Skill Tree is a modern, design-forward alternative to a traditional resume — every item on it has a verifiable credential attached.

What the Skill Tree Is

Think of it as a modern-day digital resume — but instead of just writing 'I did Python at XYZ institute', the user attaches a verified credential to that claim. Anyone who receives the skill tree can tap any node and see the blockchain-verified proof.

1	**Register as Individual** An individual (e.g. a nursing graduate, software engineer, or fresher) signs up as an individual account on TruMarkZ.
2	**Fill in the Skill Tree Template** One standard template (not multiple). The individual fills in their details across sections: Education (10th, 12th, Graduation, Post-graduation) | Courses & Certifications | Work Experience | Skills. Each item has fields: Institution name, Year, Certificate/document upload.
3	**Submit Documents for Verification** For each item on the skill tree, the individual uploads a digital copy of their certificate or document. These are verified either via API (e.g. NAD for degree) or by a human verifier (e.g. a short course certificate from a small institute).
4	**Verified Nodes Appear** Once a check is complete, that node on the skill tree gets a green verified tick and a credential ID. Tapping it opens the full verification report and blockchain proof.
5	**Share the Skill Tree** The completed skill tree can be: Shared as a link, Downloaded as a PDF (with tappable QR codes on each credential node), Sent via WhatsApp or email, Used as an attachment in a job application or college admission form.
6	**Recipient Taps Any Node** A recruiter or college admissions officer receives the skill tree. They tap 'Python — Advanced Course — XYZ Institute' and see: verified by TruMarkZ, issued date, blockchain proof. All in under 2 seconds.

Example — A Fresher's Skill Tree

- B.Com — Mumbai University — 2024 ✓Verified
- Python (Advanced) — Coursera — 2023 ✓Verified
- Data Analysis — Internshala — 2023 ✓Verified
- Internship — XYZ Company — 6 months ✓Verified
- Tally ERP Certificate — 2022 ✓Verified

Each ✓is a tappable credential with blockchain proof behind it.

Who Can Use the Skill Tree

→ Students & freshers — verified academic and course credentials
→ Nurses — qualifications + hospital experience verified
→ Engineers — degree + projects + certifications verified
→ Tradespeople — skills + work history verified
→ Anyone who wants a modern verified alternative to a traditional CV

Shared with: recruiters, colleges, hospitals, Uber, logistics companies, or any employer.

8. Our Team & How We Will Build This

The Team

Role	#	What They Build
Project Manager	1	Your primary contact. Sends a written update every Friday. Runs daily team standups. Manages the 10-week plan and flags any risks early.
Flutter Developer	1	Builds the full Android app — Organisation flow, Verifier app, Individual/Skill Tree flow, QR scanner, credential wallet, Razorpay payments, photo upload.
Web Developer	1	Builds the Organisation web portal, Super Admin panel, and all web-based dashboards. Also builds the public credential viewer page (QR scan landing page).
Backend Developer	1	Builds all server logic, connects all APIs (Aadhaar, PAN, DL, Education via NAD, Employment via EPFO, GST verification), Dhiway blockchain integration, AWS cloud setup, database, and batch processing for 10,000+ credentials.

How We Will Build Efficiently

Both the app and the backend are built in parallel — no one is waiting for the other. Here is how:

Parallel Development

→ API contracts agreed on Day 1 (Week 1)
→ App uses mock data while real APIs are built
→ Both teams merge in Week 5
→ No waiting — continuous progress

Built for Scale from Day 1

→ AWS ECS (auto-scaling containers) — handles 10 users or 100,000 users
→ Background job queue for batch credential generation (10,000+ at once)
→ Database optimised for credential lookups at high volume
→ Dhiway blockchain writes are batched — cost efficient at scale
→ All APIs have retry logic + fallback to manual queue if a government API is down

9. 10-Week Build Plan

Weeks 1–4	Weeks 5–8	Weeks 9–10
Building Everything	Testing & Connecting	Launch on Play Store

Wk	Focus	What Gets Done
1	**Foundation**	App + web structure set up. Cloud servers live. Login, registration, OTP, org verification flow. GST API integration. You get access to test environment.
2	**Org Portal**	Verification plan config, per-check cost display, Excel template generator, bulk upload, single upload form, batch creation and progress dashboard.
3	**Verifier Engine**	API connections live: Aadhaar, PAN, DL, Education (NAD), Employment (EPFO). Verifier email dispatch system. SLA monitoring + alerts. Dispute panel.
4	**Credentials**	Credential builder, all 6 templates, field selection screen, bulk generation engine, Dhiway blockchain integration, QR code system, credential wallet.
5	**Registry + Search**	Dhiway registry search, permission-based access, public viewer page, Razorpay payment for report download, invoice generation.
6	**Skill Tree + Admin**	Skill tree builder for individuals, Super Admin full panel (org approval, batch monitoring, verifier management). All screens polished.
7	**Your Testing**	You test the complete platform. QA runs formal test cases. All issues logged and fixed in the same week.
8	**Final Fixes**	All feedback applied. Security audit. App build submitted to Google Play Store. Dhiway on live network.
9	**Live Servers**	AWS production setup. Your domain connected. All APIs on live mode. Data security finalised.

Wk	Focus	What Gets Done
10	Go Live	Play Store approved. First organisations onboarded. Full team on standby 48 hours. Weekly updates begin.

10. Client Sign-Off

By signing below, the client confirms they have reviewed this document and agree that it accurately represents the scope, flows, and approach for the TruMarkZ platform.

Client Representative	Development Team Lead
Name: _____	Name: _____
Signature: _____	Signature: _____
Date: _____	Date: _____

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